

MATH 152 FINAL EXAMS PRACTICE QUESTIONS AND COMMENTS

The main essential instructions for this afternoon's exams will be as follows: Take note beforehand as I would not be clarifying instructions in the exams hall.

1. Answer **ALL** Questions by shading the correct option on the **SCANNABLE SHEET** and **CIRCLE** your answers with pen in the question paper.
2. If you believe an answer you obtain is not provided in the multiple-choice answers given, kindly **SHADE** the letter (E) on the scannable sheet and **WRITE** your answer on the question paper against the question number.
3. No communication without explicit permission from invigilation team.
4. Do not take any examination materials from the Examination Hall.

The Questions will be THIRTY (30) in all, arranged in THREE (3) parts. The following is a mock presentation of the exams' questions, with practice questions for you to consider before the exams. But before that, a few remarks:

I believe by now you are all well prepared!

- **Budget Your Time:** The questions will be MCQs (Multiple Choice Questions), 30 questions in all two hours. That means that you have on average **4 minutes** to answer each question. Endeavour therefore to time yourself per each question. Perhaps, you should attempt taking on average **3** or **3.5 minutes** per question so that you can have about half an hour at the end to go through your questions. A watch/timer at hand may be helpful to you.
- **Double-check Answers:** MCQs can be simple but delicate because you don't receive marks for attempting correctly to answer questions but unfortunately getting the final answer wrong; thus

(1) Make sure you have *understood* the question being asked. Don't *rush* reading a question.

(2) Endeavour to answer the questions you know to answer first. Usually, that means you will spend less time answering those ones and have more time (like **5** minutes) to spend on those other questions you *might* think are very difficult.

(3) Try to spot likely answers and eliminate possibly incorrect answers.

- **Look again if you think it's difficult:** I set a number of the questions to test out your *understanding* and *grasp* of the concepts we studied in class. Do not be expecting some inordinately advanced questions! Sometimes, a seemingly tricky question may actually be simple.
- **Be well-prepared:** This is certainly the crux of every exams. Know the fundamental techniques of what we did in class, and also during the revision week. Know your basic differentiation techniques, derivatives of basic functions, fundamental principles of computing limits of functions etc.
- **Be neat and avoid stray marks:** Shade your answers neatly and avoid multiple shading or stray marks that may obscure your possibly correctly shaded answers. It may happen that I may resort to a scoring machine for the scoring, in which case any multiple or stray shading will not count towards your final score tally. Remember to circle your shaded answers as well on your question sheet for the sake of future reference, if ever the need for that arises. If you feel that the answer you obtain is not part of the multiple choice answer, please re-read the instructions on the first page.

Don't worry, you'll do just fine: This is just to encourage you to relax, take your time and answer the questions (starting with those that you consider easier to you first, before coming to those you feel might be difficult).

Below are some practice questions for you, and the four parts of the exams will be as presented below.

Part I: Limits and Continuity

(1) Evaluate the limit below

$$\lim_{n \rightarrow \infty} \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n} \right)$$

- A. 1 B. 2 C. 4 D. 8

Use the function below to answer the next **two** questions.

$$f(x) = \begin{cases} 7 + 4x & \text{if } x < 1 \\ x^2 - 2 & \text{if } x \geq 1 \end{cases}$$

(2) What is $\lim_{x \rightarrow -6} f(x)$?

- A. 3 B. 17 C. 31 D. Does not exist

(3) What is $\lim_{x \rightarrow 1} f(x)$?

- A. 3 B. 16 C. 31 D. Does not exist

Use the function below to answer the next **two** questions.

$$g(x) = \begin{cases} 6x & \text{if } x \leq -4 \\ 1 - 9x & \text{if } x > 4 \end{cases}$$

(4) What is $\lim_{x \rightarrow 7} g(x)$?

- A. 2 B. 7 C. -62 D. Does not exist

(5) What is $\lim_{x \rightarrow -4} g(x)$?

- A. 2 B. 7 C. -62 D. Does not exist

(6) Which of the statements BELOW is neither true nor false?

- A. If $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exists, then f is continuous at $x = a$.
B. If f is continuous at $x = a$, then it is differentiable there.

- C. If $f'(a)$ exists, then $\lim_{x \rightarrow a} f(x)$ exists as well.
 D. If $\lim_{x \rightarrow a} f(x)$ exists, $f'(a)$ may not exist.

Part II—Differentiation and Integration

(7) Let $f(x) = \tan x$. Find $f''(x)$, the second derivative of $f(x)$.

- A. $2 \sec^2 x \tan x$ B. $2 \sec x \tan^2 x$ C. $2 \sec x \tan x$ D. $2 \sec x^2 \tan^2 x$

(8) Let $f(x) = 3^{\cos x}$. Find $f'(x)$.

- A. $(\sin x)3^{\cos x}$
 B. $(\sin x)(\ln \cos x)3^{\cos x}$
 C. $-(\sin x)(\ln 3)3^{\cos x}$
 D. $-(\sin x)3^{\cos x}$

(9) Find and simplify $f'(0)$ given

$$f(x) = (x - 1)(2x - 1)(3x - 1)(4x - 1)(5x - 1)$$

- A. 13 B. 14 C. 15 D. 16

Evaluate the following integrals. You *may* use the hints provided for some questions if you desire. Here, K is a constant of integration

(10) $\int \frac{x^3}{x^2 + 2x + 5} dx$

- A. $\frac{x^2}{2} - 2x - \ln \sqrt{x^2 + 2x + 5} - \frac{11}{2} \tan^{-1} \frac{1+x}{2} + K$
 B. $\frac{x^2}{2} + 2x - \ln \sqrt{x^2 + 2x + 5} - \frac{11}{2} \tan^{-1} \frac{1+x}{2} + K$
 C. $\frac{x^2}{2} - 2x - \ln \sqrt{x^2 + 2x + 5} + \frac{11}{2} \tan^{-1} \frac{1+x}{2} + K$
 D. $\frac{x^2}{2} - 2x - \ln \sqrt{x^2 + 2x + 5} - \frac{11}{2} \tan^{-1} \frac{1+x}{2} + K$

(11) $\int (\sin 3x \cos 4x + \sin 4x \cos 3x)$ (Hint: Use addition formula for sin)

A. $-\frac{\sin x}{7}$

B. $\frac{\sin x}{7}$

C. $-\frac{\cos x}{7}$

D. $-\frac{\cos x}{7}$

Part IV—Applications of Calculus

Use the function $y = 2x^3 - 9x^2 + 12x - 3$ to answer the next two questions

(12) What are the stationary points of y ?

- A. Only $x = 1$
- B. $x = 1$ and $x = 2$
- C. Only $x = 2$
- D. $x = 2$ and $x = -1$

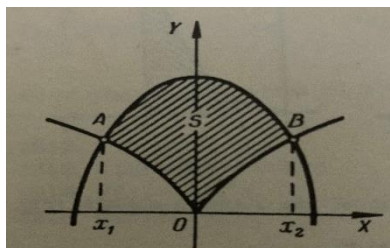
(13) The stationary points of y is/are

- A. Local maxima
- B. Local minima
- C. One is a maximum and the other a minimum
- D. Saddle points

(14) The function $y = \sin x + \cos x$ has a

- A. local maximum when $x = \frac{\pi}{4}$
- B. local minimum when $x = \frac{\pi}{4}$
- C. local maximum when $x = \frac{\pi}{2}$
- D. local minimum when $x = \frac{\pi}{2}$

(15) Find the area S between the curves $y = 2 - x^2$ and $y^3 = x^2$ below?



(Hint: Solve for the limit points of integration x_1 and x_2 , as shown in the figure).

A. $\frac{4}{15}$

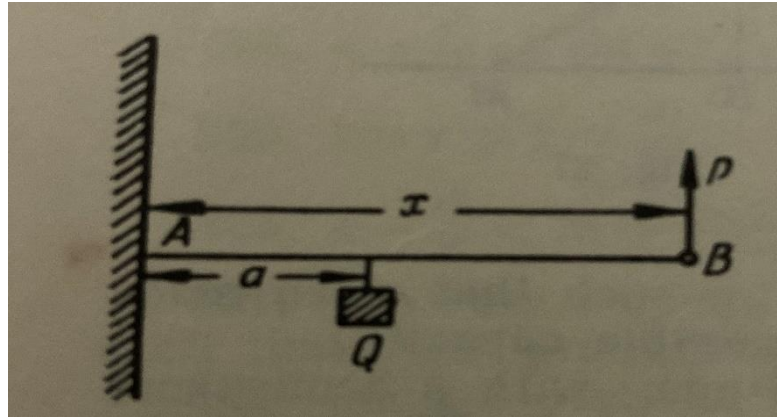
B. $\frac{3}{7}$

C. $\frac{5}{12}$

D. $\frac{4}{9}$

(16) In an engineering project, it is determined that a homogenous rod between points A and B should support a load of Q kg at a distance of a cm from point A .

The load is held by a vertical weight P (in kilogrammes) applied at point B to keep it in equilibrium. If *one centimetre* of the rod weighs w kg, determine the length x of the rod so that the force P is minimum, and calculate this minimum weight (See diagram below).



(Hint: Note that you do not require any acceleration due to gravity here because the weight here is simply the mass in kilogrammes. You must express the force P as a function of the length x of the rod. Crucial to the solution is that “equilibrium” simply means the total upward and downward weights must be the same.)

- A. length = $\sqrt{\frac{2aQ}{w}}$ and minimum weight for $P = \sqrt{2awQ}$
 B. length = $\sqrt{\frac{2Q}{aw}}$ and minimum weight for $P = \sqrt{\frac{2wQ}{a}}$
 C. length = $\sqrt{\frac{2wQ}{a}}$ and minimum weight for $P = \sqrt{\frac{2aQ}{w}}$
 D. length = $\sqrt{2awQ}$ and minimum weight for $P = \sqrt{\frac{2Q}{aw}}$

All the best today!

J. Ezearn